COURSE 2 · ALGEBRA

Algebra

Theory reader, audio edition

Expressions, equations, functions, quadratics, exponentials and logarithms, and a first look at calculus: the language every science is written in.

8 modules · equations spoken in words; numbered equations appear in the Visual Companion.

Before you start

How to use this reader

This is the audio edition. It is written to be listened to in a text-to-speech app as comfortably as read on the page. Equations are spoken in words, and each important one carries a number like this: [Equation 2.1-1]. The same number appears in the Visual Companion, where the equation is printed in ordinary notation.

Each module takes about twenty to thirty minutes to listen to. A good weekly rhythm at four hours: listen on the first day, look over the Visual Companion with a pencil, try the explorer, then spend the remaining time in the workbook.

The level sits between final-year high school and first-year university. Nothing is assumed beyond what you met at school, and it is fine if that was a long time ago. Where a first-year idea appears, it is introduced from the ground up.

In this course

2.1 Expressions and simplifying
2.2 Linear equations and inequalities
2.3 Rearranging formulas
2.4 Functions and graphs
2.5 Systems of equations
2.6 Quadratics and polynomials
2.7 Exponentials and logarithms
2.8 Calculus intuition

Algebra is the grammar of science. If Course 1 was about numbers, this course is about relationships: how one quantity depends on another, and how to turn those relationships around to find what you need. Module 2.3, Rearranging Formulas, also has its own full-length pilot materials, which go into more depth than the version here.

Module 2.1

Expressions and simplifying

Algebra uses letters to stand for numbers we don't know yet, or numbers that can change. Before solving anything you need to be fluent at tidying expressions: expanding, collecting and factorising.

By the end of this module you can

- Apply the order of operations and the rules of indices
- Expand brackets and collect like terms
- Factorise common factors and simple quadratics

Letters as numbers

An expression such as three x plus two is a recipe: take a number, triple it, add two. The letter is a placeholder. Everything you know about arithmetic still applies, including the order of operations: brackets first, then powers, then multiplication and division, then addition and subtraction.

Like terms have exactly the same letters raised to the same powers. Three x and five x are like terms and combine to eight x. Three x and three x squared are not, and stay separate, just as three apples and three oranges stay separate.

Expanding and factorising

To expand a bracket, multiply every term inside by the term outside [Equation 2.1-1]. Two brackets multiply out with each term in the first meeting each term in the second [Equation 2.1-2]. Factorising is the reverse: spotting what can be taken out as a common factor, or which pair of brackets multiplies to give a quadratic.

To factorise x squared plus five x plus six, look for two numbers that add to five and multiply to six: two and three. So it equals x plus two times x plus three. You can always check by expanding back.

Rules of indices

Powers follow a small set of rules. Multiplying powers of the same base adds the exponents; dividing subtracts them; a power of a power multiplies them [Equation 2.1-3]. A negative exponent means one over; a fractional exponent means a root, so x to the half is the square root of x [Equation 2.1-4]. These rules make scientific notation work and will be essential for exponentials in Module 2.7.

Key ideas to carry forward

- Only like terms combine
- Expanding and factorising are inverses: check one with the other
- $x^a x^b = x^{a+b}$; $(x^a)^b = x^{ab}$; $x^{-n} = 1/x^n$; x^(1/n) is the nth root

Module 2.2

Linear equations and inequalities

Solving an equation means finding the value that makes it true. For linear equations, where the unknown appears only to the first power, the method is mechanical and always works.

By the end of this module you can

- Solve linear equations, including those with brackets and fractions
- Solve linear inequalities and show them on a number line
- Interpret gradient and intercept of a straight line

Balance and undo

An equation is a balance: whatever you do to one side you do to the other. To solve two x minus three equals five, add three to both sides to get two x equals eight, then divide both sides by two: x equals four. Brackets are expanded first; fractions are cleared by multiplying everything by the common denominator.

Always check by substituting back. Two times four minus three is five. The habit costs ten seconds and catches most mistakes.

Inequalities

Inequalities use less than, greater than, and their 'or equal to' versions. They are solved like equations with one important exception: multiplying or dividing both sides by a negative number reverses the direction of the inequality [Equation 2.2-2]. Minus x greater than three becomes x less than minus three. A quick check with numbers shows why: five is greater than two, but minus five is less than minus two.

Straight lines

Every linear equation in two variables graphs as a straight line, usually written y equals m x plus c [Equation 2.2-1]. The gradient m is the rise over the run, how much y changes for each unit increase in x [Equation 2.2-3]. The intercept c is the value of y when x is zero. In science, the gradient almost always has a physical meaning: on a distance-time graph it is speed; on a mass-volume graph it is density.

Bridge from the social sciences

y = mx + c is the simple linear regression line, with the gradient as the slope coefficient and c as the intercept. Here the line usually comes from a physical law rather than a fit to noisy data, though in labs it is both.

Key ideas to carry forward

- Do the same to both sides; check by substitution
- Multiplying or dividing an inequality by a negative flips its direction
- Gradient = rise ÷ run; in science it usually means something physical

Module 2.3

Rearranging formulas

Changing the subject of a formula is the single most useful algebra skill in science. This is a short version; the full-length pilot materials for this module, with its own reader, companion, explorer and 38-problem workbook, go further.

By the end of this module you can

- Rearrange a formula by undoing operations in reverse order
- Handle roots, reciprocals and targets that appear twice
- Check a rearrangement by substitution, units and scaling

Peel from the outside

To make a quantity the subject, undo what has been done to it, in reverse order: the last operation applied is the first one undone. In the pendulum law, the period equals two pi times the square root of length over g [Equation 2.3-1]. To find g: divide by two pi, square both sides, flip both sides, multiply by L [Equation 2.3-2].

Two traps catch most students. You cannot flip individual terms: one over a plus one over b is not one over a plus b. And when the target appears twice, you must collect the terms together and factor it out before dividing.

Check every time

Substitute simple numbers into both forms. Check the units come out right. Ask whether the formula scales sensibly: if g doubles, should the period go up or down? These checks take seconds and catch almost every error.

Key ideas to carry forward

- Undo operations in reverse order, applied to both whole sides
- Flip only whole sides, never individual terms
- Target twice: collect, factor, divide

Module 2.4

Functions and graphs

A function is a rule that takes an input and gives exactly one output. Seeing functions as graphs, and knowing how a graph shifts and stretches when you change the formula, lets you read a relationship at a glance.

By the end of this module you can

- Use function notation, domain and range
- Recognise linear, quadratic, power, exponential and reciprocal graphs
- Predict how a graph moves under shifts, stretches and reflections

What a function is

A function, written f of x, maps each input x to one output. The domain is the set of allowed inputs; the range is the set of outputs that actually occur. The square root function has domain x greater than or equal to zero, because negative inputs have no real square root. The reciprocal function, one over x, excludes x equals zero.

A composite function applies one function and then another: f of g of x. An inverse function undoes a function: if f takes three to nine, its inverse takes nine back to three. Rearranging a formula, as in Module 2.3, is exactly finding an inverse function.

A gallery of shapes

Linear functions draw straight lines. Quadratics draw parabolas. Power functions like x cubed rise steeply. Reciprocal functions, one over x, have two branches that approach the axes without touching them; these approaching lines are asymptotes. Exponentials rise ever more steeply or decay towards zero. Knowing these shapes lets you guess the form of a law from a graph of data.

Transformations

Changing the formula moves the graph in predictable ways [Equation 2.4-1]. Adding a constant outside the function shifts the graph up. Replacing x with x minus h shifts it right by h, which feels backwards at first. Multiplying the whole function by a number stretches it vertically; a negative number also flips it upside down. Multiplying x inside by a number squashes the graph horizontally. The Visual Companion shows the parabola moved and flipped.

Key ideas to carry forward

- A function gives exactly one output per input
- Domain: allowed inputs. Range: resulting outputs
- $f(x - h) + k$ shifts right h and up k; $-f(x)$ reflects in the x-axis

Module 2.5

Systems of equations

When two unknowns are linked by two equations, you can find both. Systems of equations appear whenever several constraints must be met at once: balancing chemical reactions, mixing solutions, or finding where two orbits cross.

By the end of this module you can

- Solve two simultaneous linear equations by substitution and by elimination
- Interpret a solution as the intersection of two lines
- Recognise systems with no solution or infinitely many

Substitution

Use one equation to express one unknown in terms of the other, then substitute into the second. If x plus y equals eight, then y equals eight minus x. Substitute into x minus two y equals minus two: x minus sixteen plus two x equals minus two, so three x equals fourteen, and x equals fourteen thirds. Then y equals ten thirds.

Substitution is the natural method when one equation is already solved for a variable, and it also works when one equation is not linear.

Elimination

Elimination adds or subtracts multiples of the equations so that one unknown cancels [Equation 2.5-2]. If you have two x plus three y equals twelve and two x minus y equals four, subtracting the second from the first removes x: four y equals eight, y equals two, then x equals three. Choose multipliers that make the coefficients of one variable match.

What a solution means

Each linear equation is a line, and the solution is where they cross. Two parallel lines never meet, so the system has no solution. Two identical lines meet everywhere, giving infinitely many solutions. With three unknowns you need three equations, and each equation is a plane in three dimensions. This generalises to matrices, which first-year courses introduce.

Key ideas to carry forward

- Substitute when one variable is easy to isolate
- Eliminate by matching coefficients and subtracting
- Solution = intersection; parallel lines mean no solution

Module 2.6

Quadratics and polynomials

Quadratics are the simplest curves. They describe the path of a thrown ball, the area of a square, and the energy of a stretched spring, and they come with a formula that solves every one of them.

By the end of this module you can

- Solve quadratics by factorising, completing the square and the quadratic formula
- Use the discriminant to count real roots
- Sketch a parabola from its roots, vertex and intercept, and describe higher polynomials

Three ways to solve

A quadratic has the form a x squared plus b x plus c equals zero. If it factorises, set each bracket to zero: x squared minus five x plus six equals zero gives x equals two or three. Completing the square rewrites it in vertex form, which also gives the turning point. The quadratic formula always works [Equation 2.6-1]: x equals minus b plus or minus the square root of b squared minus four a c, all over two a.

The discriminant

The expression under the square root, b squared minus four a c, is called the discriminant [Equation 2.6-2]. If it is positive, there are two real roots and the parabola crosses the x-axis twice. If it is zero, there is one repeated root and the parabola just touches the axis. If it is negative, there are no real roots: the parabola misses the axis entirely. In physics, a negative discriminant often means 'this never happens', such as a ball that never reaches a certain height.

Polynomials

Polynomials add higher powers: cubics, quartics and so on. A polynomial of degree n has at most n real roots and at most n minus one turning points. The factor theorem says that if substituting x equals k gives zero, then x minus k is a factor [Equation 2.6-3]. For large x the highest power dominates, which is why a cubic eventually shoots off to plus or minus infinity.

Key ideas to carry forward

- Quadratic formula solves every quadratic
- Discriminant > 0: two roots; = 0: one; < 0: none
- If p(k) = 0 then (x − k) is a factor

Module 2.7

Exponentials and logarithms

Exponential change, where a quantity grows or shrinks by the same factor in each step, describes populations, radioactive decay, compound interest and the spread of disease. Logarithms are the tool for undoing it.

By the end of this module you can

- Describe exponential growth and decay, including half-life and doubling time
- Use the laws of logarithms and the natural logarithm
- Solve exponential equations and use log scales

Same factor, every step

Linear change adds the same amount each step; exponential change multiplies by the same factor. A population growing at three per cent a year multiplies by one point zero three every year [Equation 2.7-1]. After t years it is the starting value times one point zero three to the t. Decay works the same way with a factor less than one.

Natural processes often use the special base e, about two point seven one eight, written N equals N nought times e to the k t [Equation 2.7-2]. Positive k means growth, negative k means decay. The number e appears because it is the base for which the rate of growth exactly equals the current size, a fact that calculus makes precise.

Logarithms

A logarithm answers: what power gives this number? Log base ten of a thousand is three, because ten cubed is a thousand. The natural log, l n, uses base e. Logs undo exponentials, so to solve two to the x equals fifty, take logs of both sides.

Three laws make logs useful [Equation 2.7-3]: the log of a product is the sum of the logs; the log of a quotient is the difference; and the log of a power brings the exponent down in front. That last law is what lets you solve for an exponent.

Half-life, doubling time and log scales

The time to halve is the half-life; the time to double is the doubling time. Both are constant for an exponential process [Equation 2.7-4]: a half-life of five years means half remains after five years, a quarter after ten, an eighth after fifteen. The rule of seventy gives a quick doubling time: seventy divided by the percentage growth rate.

Log scales compress huge ranges. The pH scale, the decibel scale and the Richter scale all go up by one for each tenfold increase. A pH three solution is ten times more acidic than pH four, and a hundred times more than pH five.

Bridge from the social sciences

The log transforms you have used in regression, log-linear and log-log models, are this module applied to data. Exponential growth with a constant percentage rate is the same idea as continuous compounding.

Key ideas to carry forward

- Exponential: multiply by the same factor each step
- Logs undo exponentials; $\log(a^b) = b \log a$
- Half-life and doubling time are constant; $t_{1/2} = \ln 2 / k$
- Log scales: +1 means ×10

Module 2.8

Calculus intuition

Calculus is the mathematics of change. It has two halves: differentiation, which finds rates of change, and integration, which adds up accumulated change. You need the ideas, and a little of the technique, for first-year physics, chemistry and biology.

By the end of this module you can

- Interpret a derivative as a gradient and a rate of change
- Differentiate polynomials using the power rule
- Interpret an integral as an area and an accumulation, and integrate simple powers

Rates of change

The gradient of a straight line tells you how fast y changes with x. For a curve, the gradient changes from point to point. The derivative is the gradient at a single point, found by taking the gradient between two very close points and letting the gap shrink to nothing [Equation 2.8-1]. Written d y by d x, it reads as 'the rate of change of y with respect to x'.

Velocity is the derivative of position with respect to time; acceleration is the derivative of velocity. The rate of a chemical reaction is the derivative of concentration. The growth rate of a population is the derivative of its size. Every 'rate' in science is a derivative.

The power rule

For powers of x, differentiation has a simple rule [Equation 2.8-2]: bring the power down in front and reduce it by one. The derivative of x squared is two x; of x cubed is three x squared; of a constant is zero, because a constant does not change. Derivatives of sums are sums of derivatives. Two more results are worth knowing: the derivative of e to the x is e to the x, and of sine is cosine [Equation 2.8-3].

Where the derivative is zero, the curve is momentarily flat: a peak, a trough or a pause. This is how calculus finds maxima and minima, such as the angle that throws a ball furthest.

Accumulation and area

Integration runs the other way. If you know a car's speed at every moment, the distance travelled is the area under the speed-time graph. Adding up many thin strips and letting their width shrink gives the integral [Equation 2.8-4]. The fundamental theorem of calculus says integration and differentiation are inverse operations, so to integrate x to the n you raise the power by one and divide by the new power [Equation 2.8-5].

You will meet integrals as areas under graphs in physics, as total work done by a varying force, and as total drug exposure in pharmacology. For this course, the intuition matters more than the technique.

Bridge from the social sciences

Marginal effects in economics are derivatives, and cumulative totals are integrals. If you have read 'the marginal cost', you have read a derivative.

Key ideas to carry forward

- Derivative = instantaneous rate of change = gradient of the tangent
- Power rule: $d/dx(x^n) = nx^{n-1}$
- Derivative zero at peaks and troughs
- Integral = area under the curve = accumulated total